

- 7 A particle P is projected from a point O on a horizontal plane and moves freely under gravity. The initial velocity of P is 100 ms^{-1} at an angle θ above the horizontal, where $\tan \theta = \frac{4}{3}$. The two times at which P 's height above the plane is $H \text{ m}$ differ by 10 s .

(a) Find the value of H .

[5]

[illegible]

- (b)** Find the magnitude and direction of the velocity of P one second before it strikes the plane. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.